particularly challenging. It turned out, however, that a computer can defeat the world chess champion far more easily than replace a kitchen porter. Sure, automatic dishwashers have been around for decades, but even our most sophisticated robots still lack the intricate skills needed to pick up dirty dishes from the tables of a busy restaurant, place the delicate plates and glasses inside the automatic dishwasher and take them out again.

Similarly, to judge by their pay, you could assume that our society appreciates doctors more than nurses. However, it is harder to automate the job of nurses than the job of at least those doctors who mostly gather medical data, provide a diagnosis and recommend treatment. These tasks are essentially pattern recognition, and spotting patterns in data is one thing AI does better than humans. In contrast, AI is far from having the skills necessary to automate nursing tasks such as replacing bandages on an injured person or giving an injection to a crying child.[10] These two examples don't mean that dishwashing or nursing could never be automated, but they indicate that people who want a job in 2050 should perhaps invest in their motor and social skills as much as in their intellect.

Another common but mistaken assumption is that creativity is unique to humans so it would be difficult to automate any job that requires creativity. In chess, however, computers are already far more creative than humans. The same may become true of many other fields, from composing music to proving mathematical theorems to writing books like this one. Creativity is often defined as the ability to recognise patterns and then break them. If so, then in many fields computers are likely to become more creative than us, because they excel at pattern recognition.[11]

A third mistaken assumption is that computers couldn't replace humans in jobs requiring emotional intelligence, from therapists to teachers. This assumption depends, however, on what we mean by emotional intelligence. If it means the ability to correctly identify emotions and react to them in an optimal way, then computers may well outperform humans even in emotional intelligence. Emotions too are patterns. Anger is a biological pattern in our body. Fear is another such pattern. How do I know if you are angry or fearful? I've learned over time to recognise human emotional patterns by analysing not just the content of what you say but also your tone of voice, your facial expression and your body language.[12]

AI doesn't have any emotions of its own, but it can nevertheless learn to recognise these patterns in humans. Actually, computers may outperform